

IX. Mini Project – Sleep Analytics

GITHUB LINK : <https://github.com/alwinabishekst/FDVA---SLEEP-ANALYTICS-MINI-PROJECT.git>

AIM:

To clean, process, and visualize global sleep health, lifestyle, and disorder datasets using Python, and build static visualizations to derive insights on sleep patterns, quality, and disturbance trends across age groups and seasons.

DOMAIN:

Health & Sleep Analytics

PROBLEM STATEMENT:

With rising awareness of sleep health and its link to overall well-being, individuals and researchers lack a unified visual dashboard that compares sleep quality, sleep stage composition, disturbance frequency, and disorder prevalence across age groups and seasons. There is a need for a data-driven visualization system to identify at-risk groups, seasonal patterns, and lifestyle factors affecting sleep.

OUTPUT:

Python:

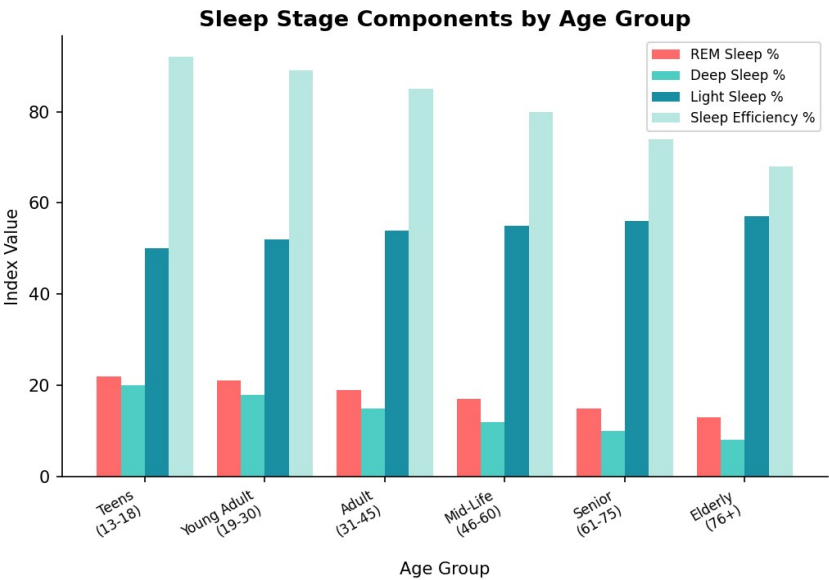


Fig 1. Sleep Stage Components (REM, Deep, Light, Efficiency) by Age Group

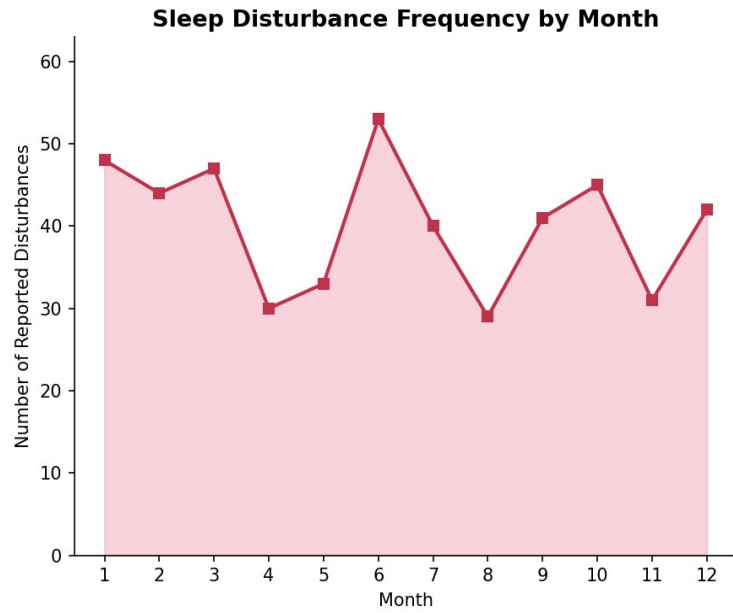


Fig 2. Sleep Disturbance Frequency by Month

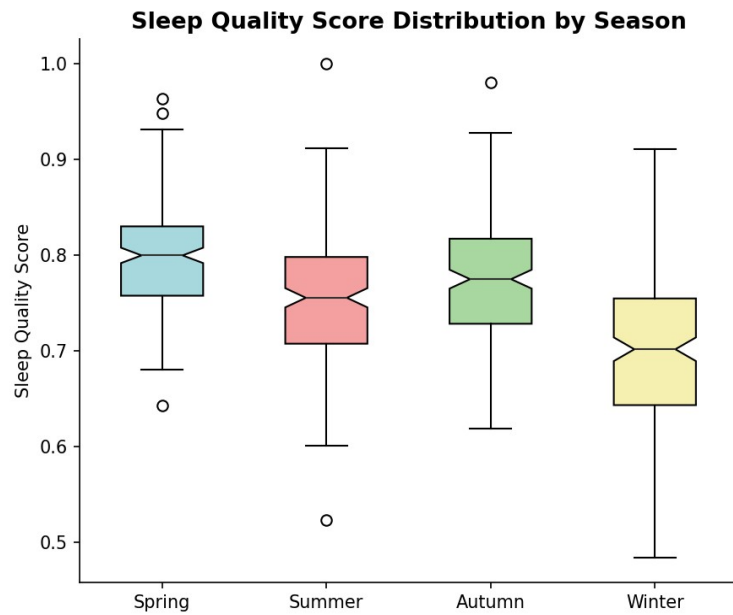


Fig 3. Sleep Quality Score Distribution by Season

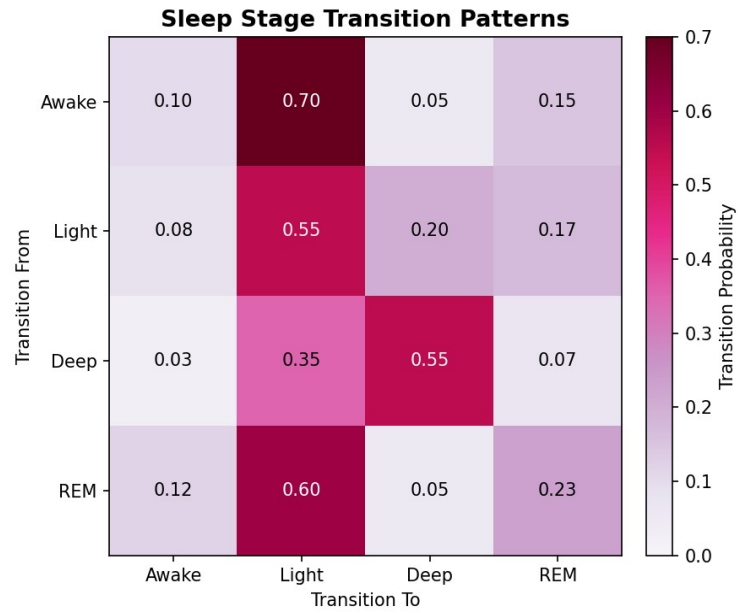


Fig 4. Sleep Stage Transition Patterns (Markov Transition Probabilities)

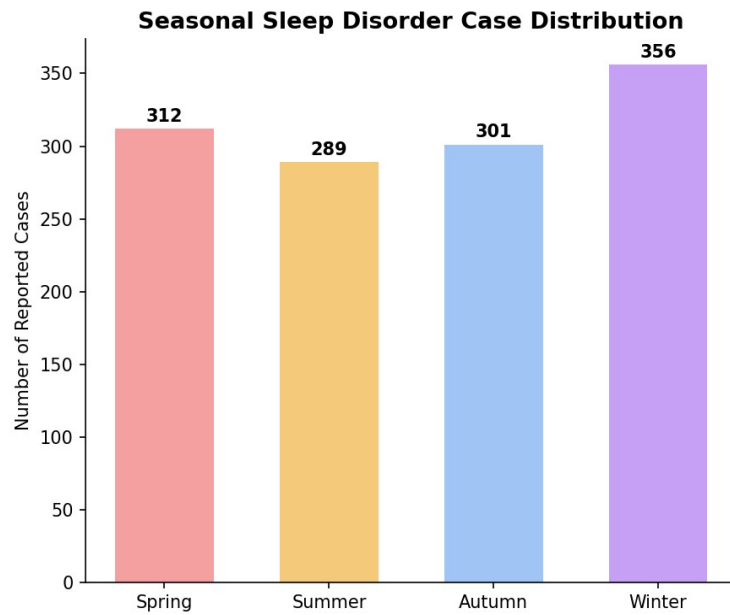


Fig 5. Seasonal Sleep Disorder Case Distribution

Sleep Disorder Category Share

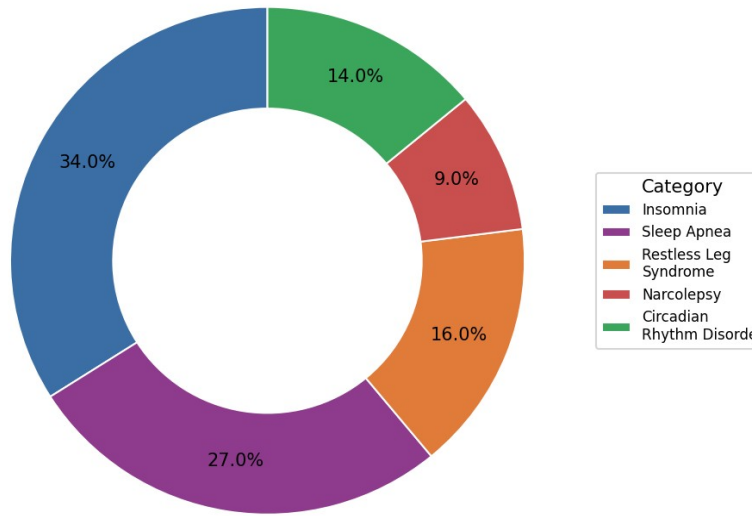


Fig 6. Sleep Disorder Category Share

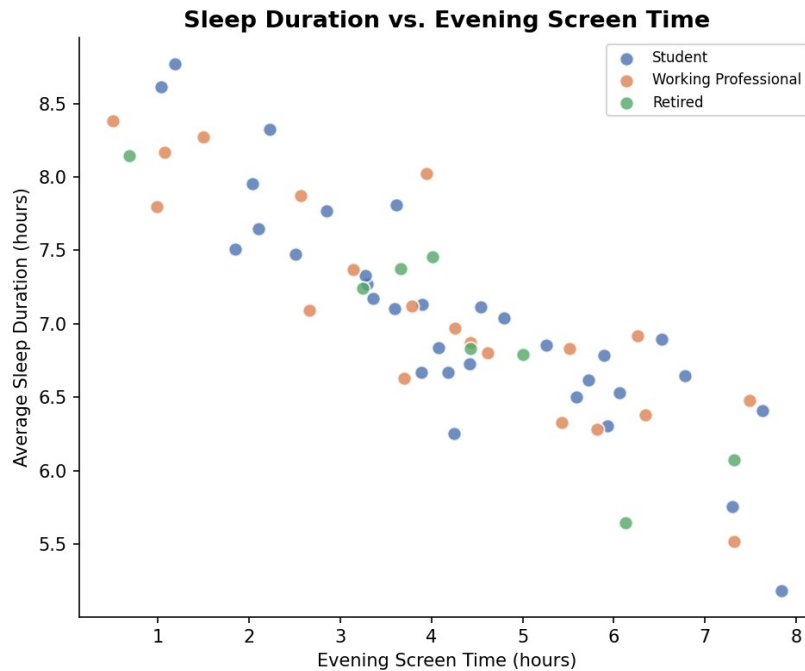


Fig 7. Sleep Duration vs. Evening Screen Time, by Occupation Group

INSIGHTS:

REM and deep sleep percentages decline steadily with age, while light sleep share and sleep-onset difficulty increase, and overall sleep efficiency drops from about 92% in teenagers to under 70% in the elderly.

Sleep disturbances peak in June and dip in April and August, suggesting a seasonal, possibly temperature- or daylight-linked pattern rather than a purely random distribution.

Winter shows the lowest median sleep quality score and the widest spread, while spring shows the highest and most consistent scores.

The transition heatmap shows sleep is dominated by frequent shifts into light sleep from every stage, while deep sleep is the most 'sticky' stage once entered (55% self-transition probability).

Insomnia and sleep apnea together account for over 60% of reported sleep disorder cases, making them the priority targets for intervention.

Evening screen time shows a clear negative relationship with sleep duration across all occupation groups, with students showing the steepest sensitivity.

RESULT:

Thus, the given program was written and executed successfully.